

complexity of your algorithm?

8. You are given an array of N integers and told that the array is mostly sorted, with only k elements not being in their correct sorted position in the array, and with k being very small compared to N . What is the sorting algorithm you would use to completely sort the array?
9. You are given an array of N integers. You are asked to find the maximum and minimum elements in each of the sub-sequences of length K of the array, where K is an arbitrary integer. (a) Given an array of N integers and a value K , how many sub-sequences of length K are possible? (b) What is the complexity of computing the maximum and minimum for each of the sub-sequences of length K ? How would you solve this problem in polynomial time, for any arbitrary value of K ?
10. You are given a list of N integers, where the numbers can only be positive. You are now being asked to write code to create two lists of size $N/2$ such that the sum of the elements of the two sub-lists are equal, if at all it is possible to create such sub-lists. Your function should return the sum of the sub-list elements if it is possible to split the original list into two such sub-lists, or return 0 if that is not possible. What is the time complexity of your function? (Note that you can move the elements of the list.) You are now told that the original list can either contain positive or negative integers, and are told to solve the above problem. What are the changes you would make to your earlier solution?
11. Consider the problem given in (10). Now, instead of splitting the list into two sub-lists such that the elements of both are equal, you are asked to split the list at any arbitrary index such that the difference of the sum of the two sub-lists is the minimum among all possible splits of the original list. Unlike problem (10), you cannot move around the elements of the list. What is the time and space complexity of your function?
12. You are given two sentences $S1$ and $S2$, each containing N and M words, respectively. You are now asked to convert $S1$ into $S2$ by either deleting a word or adding a new word. Modifying word1 to word2 can be considered as word deletion, followed by word addition. Each insert operation has a cost of +2 and each delete operation has a cost of +1. Given two arbitrary sentences $S1$ and $S2$, the edit distance of ($S1$, $S2$) is the minimum total cost of operations needed to convert $S1$ into $S2$. You are asked to write a function to compute the edit distance. What is the

time and space complexity of your function?

13. You are given a long piece of text T and three sub-strings $S1$, $S2$ and $S3$. You are asked to find the shortest sub-string X of T such that X contains $S1$, $S2$ and $S3$ as sub-strings. What is the time and space complexity of your solution?
14. You are given an array of N integers (where N can be of the order of millions) which has been sorted. Given an arbitrary value K , you are asked to find out how many times it repeats in the array? Can you write a C function to do this in $O(\log N)$ time complexity?
15. Consider the following code snippet:

```
int main(int argc, char *argv[])
{
    void *my_mem = 0;
    int alloc_blocks = 0;
    while(1)
    {
        my_mem = (void *) malloc(1024 * 1024);
        if (!my_mem) break;
        memset(my_mem, 1, (1024*1024));
        printf("allocated memory blocks is %d MB\n", ++alloc_blocks);
    }
    exit(0);
}
```

Can you explain what would happen when you compile and run the program?

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming! 

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